

B. Tech Degree VIII Semester Examination, April 2008

ME 802 MACHINE DESIGN II *(2002 Scheme)*

Time : 3 Hours

Maximum Marks : 100

(Design data book is permitted)
(Assume reasonable data wherever necessary)

- I. (a) What are the desirable properties of materials used for brake lining? (4)
(b) Differentiate between uniform pressure and uniform wear theories. (4)
(c) For an automotive type internal expanding double shoe brake, the following data are given :
Drum rotation : clockwise $\theta_1 = 15^\circ$, $\theta_2 = 105^\circ$
Drum radius = 125 mm, $\mu = 0.25$, $b = 65$ mm, $P_{\max} = 0.6$ N/mm²
Vertical distance from the drum centre to the fulcrum pin = 90 mm
Vertical distance from the drum centre to the line of actuating force = 90 mm
Horizontal distance of fulcrum pins from centre of drum = 40 mm
Determine (i) Actuating force (ii) Braking torque of the shoe brake. (17)
- OR**
- II. (a) Why tight side of the belt should be at the bottom side of the pulleys? (4)
(b) Explain the different chain drive failures. (4)
(c) Design a V belt drive for the following data :
Drive : AC motor, Operating speed is 1440 rpm and operates over 10 hours. The equipment driven is a compressor which runs at 900 rpm and the required power transmission is 20 KW. (17)
- III. (a) What are the contributory factors for dynamic loading of spur gearing? (4)
(b) Explain the different modes of gear failure. (4)
(c) A pair of spur gears having $14\frac{1}{2}^\circ$ involute full depth teeth is to transmit 12 KW at 300 rpm of the pinion. The velocity ratio is 3:1. The static strengths of cast iron gear and steel pinion are 60 MPa and 105 MPa respectively. Determine the module, face width and pitch diameters of the gears. Also check the gears for wear. Number of teeth on the pinion is 16. (17)
- OR**
- IV. (a) Explain the term formative number of tooth in helical gears. (4)
(b) 'Worm gearing is not interchangeable'. Explain. (4)
(c) A pair of CI bevel gears connect two shafts at right angle. The pitch diameters of pinion and gear are 80 mm and 120 mm respectively. The tooth profiles of the gears are of $14\frac{1}{2}^\circ$ composite form. The static strength for both the gears is 56 MPa. The pinion transmits 2.5 KW at 1200 rpm. Determine the module for the gears from the standpoint of strength and check the design for wear. Take surface endurance limit as 620 MPa and $E = 0.84 \times 10^5$ MPa. (17)
- V. (a) Explain the hydrodynamic theory of lubrication. (5)
(b) Distinguish between Hydrodynamic bearings and Hydrostatic bearings. (5)

(Turn Over)

- (c) A full journal bearing of 0.10 m diameter by 0.15 m long supports a radial load of 5500 N. The shaft speed is 500 rev/min. The room temperature is 32°C and the surface of the bearing is to be limited to 63°C. Select a suitable oil to satisfy the above requirements, if the bearing is well ventilated and no artificial cooling to be used. Assume $r/c = 1000$. (15)

OR

- VI. (a) What are the advantages and disadvantages of rolling contact bearings? (5)
 (b) What are the different factors governing the selection of bearings? (5)
 (c) A simply supported shaft, diameter 50 mm, on bearing supports carries a load of 10 kN at its centre. The axial load on bearings is 3 kN. The shaft speed is 1440 rpm. Select a bearing for 1000 hours of operation. (15)

- VII. (a) How working drawings are prepared? With examples show how tolerance and finish are represented in working drawing. (10)
 (b) What are the general design recommendations for forging? (8)
 (c) Enumerate the general design recommendations for turned parts. (7)

OR

- VIII. (a) Explain the terms limits and fits. (5)
 (b) Give the design recommendations for
 (i) Rolled parts (10)
 (ii) Machined round holes (5)
 (iii) Welded joints. (5)

